TUTORIAL WORKBOOK

24MT2012 - MATHEMATICAL OPTIMIZATION

Koneru Lakshmaiah Educational Foundation

Student Id:

Student Name:

Academic Year: 2025-26

TUTORIAL WORKBOOK

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STUDENT NAME	
REG.NO	
YEAR	
SEMESTER	
SECTION	
FACULTY	

ORGANISATION OF STUDENT TUTORIAL WORKBOOK

The Tutorial framework includes a creative element but shifts the time-intensive aspects outside of the One-Hour closed Tutorial period. Within this structure, each Tutorial includes three parts: Pre-Tutorial, In-Tutorial, Post-Tutorial.

a **Pre-Tutorial**: Pre-tutorials help students in obtaining a mathematical model for a real-life problems and familiarize themselves with the content and structure of the upcoming tutorial.

b **In-Tutorial**: The In-Tutorial section takes place during the actual Tutorial period. The First hour of the Tutorial period can be used to resolve any problems the students might have experienced. The intent is to give constructive feedback so that students leave the Tutorial hour with working and solving problems - a significant accomplishment on their part.

c **Post-Tutorial**: The last phase of each Tutorial is a homework assignment that is done following the Tutorial period. In the Post-Tutorial, students analyze the efficiency or utility of a given system call. Each Post-Tutorial exercise should take roughly 120 minutes to complete.

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2025-26 EVEN SEMESTER TUTORIAL CONTINUOUS EVALUATION

SL.NO	DATE	TUTORIAL TITLE	PRE-TUTORIAL (10M)	IN-TUTORIAL (20M)	POST-TUTORIAL (15M)	VIVA-VOCE (5M)	TOTAL (50M)	FACULTY SIGNA-TURE
1		Formulation of LPP and Graphical Method						
2		Simplex Method and Principle of Duality						
3		Transportation Problem						
4		Gomory's Cut-plane Method						
5		Branch and Bound Method						
6		Constrained optimization						
7		Quadratic Optimization using Wolfe's method						
8		Quadratic Optimization using Beale's Method						
9		Geometric Programming Problem						
10		Genetic Algorithm						
11		Ant Colony Optimization (ACO)						
12		Particle Swarm Optimization (PSO)						

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Tutorial 1

Formulation of Linear Programming Problem (LPP) and Graphical method

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Date of the Session: .....

Learning outcomes:

- Understanding the process of formulating a given problem in linear form.
- This requires defining the decision variables of the problem, establishing inter-relationship between them and formulating the objective function and constraints.
- Understanding the process of solving a given LPP using graphs will be discussed.

1.1 PRE-TUTORIAL

1. What is Linear Programming Problem (LPP)?

2. Define the following terminology in LPP:
   (a) Objective function
   (b) Constraints
   (c) Decision variables
   (d) Non-Negativity Conditions

3. Enumerate the steps involved in Formulation of LPP.

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4. Which type of L.P.P. can be solved using graphical method?

1.2 IN-TUTORIAL

1. A hotel has requested a manufacturer to produce pants and jackets for their boys. For materials, the manufacturer has $750m^2$ of cotton textile and $1000m^2$ of silk. Every pair of pants (1 unit) needs $2m^2$ of silk and $1m^2$ of cotton. Every jacket needs $1.5m^2$ of cotton and $1m^2$ of silk. The price of the pants is fixed at \$50 and the jacket, \$40. What is the number of pants and jackets that the manufacturer must give to the hotel so that these items obtain a maximum sale? Formulate the problem using mathematical modeling of LPP and define the objective function?
Solution:

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2. A transport company has two types of trucks, Type A and Type B. Type A has refrigerated capacity of $20m^3$ and a non-refrigerated capacity of $40m^3$ while Type B has refrigerated capacity of $30m^3$ and non-refrigerated capacity of $30m^3$. A grocer needs to hire trucks for the transport of $3000m^3$ of refrigerated stock and $4000m^3$ of non-refrigerated stock. The cost per kilometer of a Type A is \$30 and \$40 for Type B. How many trucks of each type should the grocer rent to achieve the minimum total cost? Formulate the problem using mathematical modeling of LPP and define the objective function?
Solution:

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3. A hotel has requested a manufacturer to produce pillows and blankets for their room service. For materials, the manufacturer has $750m^2$ of cotton textile and $1000m^2$ of silk. Every pillow needs $2m^2$ of cotton and $1m^2$ of silk. Every blanket needs $2m^2$ of cotton and $5m^2$ of silk. The price of the pillow is fixed at \$5 and the blanket is fixed at \$10.What is the number of pillows and blankets that the manufacturer must give to the hotel so that these items obtain a maximum sale? Formulate the LPP and solve the LPP using Python PuLP.
Solution:

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4. An Industry makes two items of P and Q by using two devices X and Y. Processing time requires 50 hrs for item P on device X and 30 hrs requires on device Y. Processing time requires 24 hrs for item Q on device X and 33 hrs requires on device Y. At starting of the current week, 30 pieces of A and 90 pieces of B are available. Processing time that is available on device X is predict to be 40 hrs and on device Y is predict to be 35 hrs. Demand for P in the current week is predict to be 75 pieces and for Q is predict to be 95 pieces. Industry policy is to maximize the combined sum of the pieces of P and the pieces of Q in stock at the end of the week. Formulate the problem of deciding how much of each item to make in the current week as a linear program. Obtain the solution using graphical method. Also, solve the LPP using Python PuLP
Solution:

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5. Consider the following linear programming problem

$$\begin{aligned} \text{Maximize:} \quad & P = 7x + 12y \\ & 2x + 3y \leq 6 \\ & 3x + 7y \leq 12 \end{aligned}$$

Obtain the solution using graphical method.

Solution:

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1.3 CASE STUDY

1.3.1 Task Assignment Optimization

Consider a manufacturing facility with 4 tasks and 3 machines. The processing times and machine capacities are given below.

Table 1.1: Processing time

Task	**Machine 1**	**Machine 2**	**Machine 3**
Task 1	5	7	3
Task 2	2	4	6
Task 3	8	3	2
Task 4	6	2	4

Table 1.2: Processing time

Machine	**Capacity**
Machine 1	15
Machine 2	10
Machine 3	20

Write Python code to solve this optimization problem using the PuLP library to minimize the total completion time.

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1.3.2 Software Development Project Scheduling

Consider a software development project with five tasks that need to be scheduled. Each task has a specific duration and may depend on the completion of other tasks. There are constraints on the availability of developers. The objective is to minimize the total project completion time.

Table 1.3: Task Durations and Dependencies

Task	**Duration (days)**	**Dependencies**
A	5	None
B	8	None
C	6	A
D	4	B
E	7	C, D

Table 1.4: Developer Availability

Developer	**Available Days**
Developer 1	10 days
Developer 2	15 days

Minimize the total project completion time, subject to:

- Task durations.
- Task dependency constraints (i.e., a task cannot start until all its dependencies are completed).
- Developer availability limits.

Write Python code to solve this optimization problem using the PuLP library.

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Tutorial 2

Simplex method and Principle of Duality

Date of the Session: .....

Learning outcomes:

- Understanding the key terms: feasible solution, feasible region and optimal solution.
- Understanding the limitations of graphical method and introduce the Simplex algorithm.
- Understand that an LPP consists of more number of constraints as compared to number of decision variables.
- Understanding the computational procedure can be considerably reduced by converting the LPP into a form called as DUAL and then solving it.

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2.1 PRE-TUTORIAL

1. Which type of L.P.P. can be solved using Simplex method?

2. What do you mean by feasible region, feasible solution and optimal solution?

3. State the general rules for formulating a dual LPP from its primal?

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2.2 IN-TUTORIAL

1. Consider the following linear programming problem

$$\begin{aligned} \text{Maximize:} \quad & P = 7x + 12y \\ & 2x + 3y \leq 6 \\ & 3x + 7y \leq 12 \end{aligned}$$

Set up the Initial Simplex Tableau and obtain the solution.

Solution:

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2. Find the dual problem for the given LPP model.

$$\begin{aligned} \text{Minimize}: \; & C = 5x_1 + 2x_2 \\ \text{Subject to}: \; & x_1 + 3x_2 \geq 15, \\ & 2x_1 + x_2 \geq 20, \\ & x_1, x_2 \geq 0 \end{aligned}$$

Solution:

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3. State the dual for the following LPP and hence solve LPP.

$$\begin{aligned} \text{Minimize}: \ C &= 21x_1 + 50x_2 \\ \text{Subject to}: \ & 2x_1 + 5x_2 \geq 12, \\ & 3x_1 + 7x_2 \geq 17, \\ & x_1, x_2 \geq 0 \end{aligned}$$

Solution:

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2.3 POST-TUTORIAL

1. Solve LPP using the simplex method.

$$
\begin{aligned}
&\text{Maximize:} \quad Z = 3x_1 + 5x_2 \\
&\text{Subject to:} \\
&\quad 3x_1 + 2x_2 \leq 18 \\
&\quad x_1 \leq 4 \\
&\quad 2x_2 \leq 12 \\
&\quad x_1, x_2 \geq 0
\end{aligned}
$$

Solution:

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2. Find its dual and obtain the optimal solution for minimization problem.

$$\begin{aligned} \text{Minimize}: \ C &= 16x_1 + 8x_2 + 4x_3 \\ \text{Subject to}: \ & 3x_1 + 2x_2 + 2x_3 \geq 16, \\ & 4x_1 + 3x_2 + x_3 \geq 14, \\ & 5x_1 + 3x_2 + x_3 \geq 12, \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Solution:

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3. A producer of Healthy food makes two important and secret ingredients that goes into their human food, named as a Healthy Man and Common Man. Each kg of Healthy Man contains 300 g of vitamins, 400 g of protein, and 100 g of carbs. Each kg of common man contains 100 g of vitamins, 300 g of protein, and 200 g of carbs. Guidelines for minimum nutritional that require a mixture made from these ingredients contain at least 900 g of vitamins, 2400 g of protein, and 800 g of carbs. Healthy Man costs \$2 per kg to produce and Common Man costs \$1.25 per kg to produce. Find the number of kgs of each ingredient that should be produced in order to minimize cost. Obtain its Dual and solve.

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Tutorial 3

Transportation problem

Date of the Session: .....

Learning outcomes:

- Understanding the problem of transporting/shipping the commodities from the industry to the destinations with the least possible cost while satisfying the supply and demand limits.

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3.1 PRE-TUTORIAL

1. List out different types of transportation methods.

2. List out the steps involved in solving the Transportation problem using North-West corner rule?

3. List out the steps involved in solving the Transportation problem using Row Minimum method.

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3.2 IN-TUTORIAL

1. Luminous lamps have three factories - F1, F2, and F3 with production capacity 30, 50, and 20 units per week respectively. These units are to be shipped to four warehouses W1, W2, W3, and W4 with requirement of 20, 40, 30, and 10 units per week respectively. The transportation costs (in Rs.) per unit between factories and warehouses are given below.
   Solve Transportation problem using NW corner rule.

Factory	Warehouse				Supply
	W1	W2	W3	W4	
F1	1	2	1	4	30
F2	3	3	2	1	50
F3	4	2	5	9	20
Demand	20	40	30	10	

Solution:

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2. The Ushodaya departmental store has three plants located throughout a state with production capacity 80, 60 and 70 kilo grams of rice. Each day the firm must furnish its four retail shops R1, R2, R3, R4 with at least 40, 60, 50, and 60 gallons respectively. The transportation costs (in Rs.) are given below. Solve Transportation problem using Row Minimum method.

Store	Retailshop				Supply
	1	2	3	4	
1	3	5	7	6	80
2	2	5	8	2	60
3	3	6	9	2	70
Demand	40	60	50	60	

Solution:

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3.3 Case study

3.3.1 Optimize the transportation cost

Consider a scenario where there are 3 suppliers and 4 consumers. The transportation costs from each supplier to each consumer are as follows:

Table 3.1: Transportation Costs

	Consumer 1	Consumer 2	Consumer 3	Consumer 4
Supplier 1	4	6	8	7
Supplier 2	2	5	3	1
Supplier 3	7 4	6	9	

The supply capacities of the suppliers and the demand requirements of the consumers are as follows:

Table 3.2: Capacities and Demands

Supply	Capacity
Supplier 1	20
Supplier 2	30
Supplier 3	50

Table 3.3: Capacities and Demands

Demand	Requirement
Consumer 1	25
Consumer 2	35
Consumer 3	30
Consumer 4	40

Write Python code to solve to find the Minimized the total transportation cost to this transportation logistics problem

Solution:

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3.3.2 Optimize the transportation cost

Consider another scenario with 2 suppliers and 3 consumers. The transportation costs and capacities/demands are as follows:

Table 3.4: Transportation Costs

	Consumer 1	Consumer 2	Consumer 3
Supplier 1	3	5	2
Supplier 2	4	6	8

Table 3.5: Capacities and Demands

Supplier	Capacity
Supplier 1	30
Supplier 2	40

Table 3.6: Demand Requirement

Demand	Requirement
Consumer 1	20
Consumer 2	25
Consumer 3	30

Write Python code to Minimize the total transportation cost for the transportation logistics problem.
Solution:

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Tutorial 4

Gomory's Cut-Plane Method

Date of the Session: .....

Learning outcomes:

- If the optimal solution is integers, then problem is solved. Otherwise, add Gomory's constraint (cut) is added to optimal solution Now new problem is solved using dual simplex method The method terminates as soon as optimal solution become integers.

4.1 PRE-TUTORIAL

1. Enumerate the steps involved in Gomory's cutting plane method.

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4.2 IN-TUTORIAL

1. Find solution using integer simplex method (Gomory's cutting plane method)

$$\begin{aligned}
&\text{Maximize} Z = x_1 + x_2 \\
&\text{subject to} \\
&\qquad 3x_1 + 2x_2 \leq 5 \\
&\qquad x_2 \leq 2 \\
&\qquad \text{and } x_1, x_2 \text{ are non-negative integers}
\end{aligned}$$

Solution:

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4.3 POST-TUTORIAL

1. Consider the following linear integer programming problem

$$\text{Maximize} Z = 14x_1 + 16x_2$$

Subject to the constraints

$$4x_1 + 3x_2 \leq 12$$
$$6x_1 + 8x_2 \leq 24$$

and $x_1, x_2 \geq 0$ and are integers.

Solve the problem by Gomory's cutting plane method

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Tutorial 5

Branch and Bound Method

Date of the Session: .....

Learning outcomes:

- Understanding to do the process of solving branch and bound method.
- Understanding to do the process of Gomory's cutting plane method.

5.1 PRE-TUTORIAL

1. What is branch and bound Technique?

2. Which strategy can be used to solve branch and bound problem?

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5.2 IN-TUTORIAL

1. Discrete Optimization using Cutting Plane method Solve the integer programming problem

$$
\begin{aligned}
&\text{Maximize:} \quad Z = 3x_1 + x_2 + 3x_3 \\
&\text{Subject to:} \\
&\quad -x_1 + 2x_2 + x_3 \leq 4 \\
&\quad 2x_2 - \frac{3}{2}x_3 \leq 1 \\
&\quad x_1 - 3x_2 + 2x_3 \leq 3
\end{aligned}
$$

Where $x_1, x_2, x_3 \geq 0$ and integer. Get the optimal solution as an integer value using Gomory's cutting plane method.
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2. Use Branch and Bound method to

$$\begin{aligned}
&\text{Maximize:} \quad Z = 3x_1 + 5x_2 \\
&\text{Subject to:} \\
&\quad 2x_1 + 4x_2 \leq 25 \\
&\quad x_1 \leq 8 \\
&\quad 2x_2 \leq 10
\end{aligned}$$

Where x_1, x_2 are non-negative integers.

Solution:

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5.3 POST-TUTORIAL

1. Explain Gomory's method for solving an Integer Programming Problem and hence solve the following,

$$\begin{aligned}
&\text{Maximize:} \quad Z = 2x_1 + 6x_2 \\
&\text{Subject to:} \\
&\quad 3x_1 + x_2 \leq 5 \\
&\quad 4x_1 + x_2 \leq 9
\end{aligned}$$

Where $x_1, x_2 \geq 0$ and are integers.

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Tutorial 6

Constrained optimization

Date of the Session: .....

Learning outcomes:

- Understanding to do the process of gradient descent method.
- Understanding to do the process of Newtons method.

6.1 PRE-TUTORIAL

1. Explain the concept of gradient descent, including the mathematical rule used to iteratively adjust parameters.

2. Describe how Newton's method operates as an optimization algorithm, and state the specific formula governing its parameter adjustments

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6.2 IN-TUTORIAL

1. Given the following set of ordered pairs (3,8), (9,6), (5,4), (3,2), and assume that the relationship between x,y is approximately linear(y=mx+c). Your task is to optimize the parameters m (slope) and c (intercept) by minimizing the Mean Squared Error (MSE) cost function using the gradient descent algorithm.

Solution:

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6.3 POST-TUTORIAL

1. Apply Newton's method, find the minimizer of $f(x) = 1/2x^2 - sinx$ The initial value is x(0)=0.5 the required accuracy is10^-5.
   Solution:

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2. How does the learning rate affect the convergence of gradient descent? Demonstrate with an example. $f(x) = (x-3)^2$ alpha equal to 0.1.
   Solution:

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3. Why is the Conjugate Gradient method more efficient than standard gradient descent for solving large symmetric positive-definite systems?

Solution:

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Tutorial 7

Quadratic Optimization using Wolfe's Method

Date of the Session: .....

Learning outcomes:

- Understanding conic duality and finding the duality of given problem.
- Understanding Lagrange function.
- Understanding the process of Wolfe's methods.

7.1 PRE-TUTORIAL

1. Outline the steps involved in Wolfe's method.

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7.2 IN-TUTORIAL

1. Use Wolfe's method for solving quadratic programming problem.

$$\begin{aligned}
&\text{Maximize:} \quad Z = 4x_1 + 6x_2 - 2x_1^2 - 2x_1x_2 - 2x_2^2 \\
&\text{Subject to:} \\
&\quad x_1 + 2x_2 \leq 2 \\
&\quad x_1, x_2 \geq 0
\end{aligned}$$

Solution:

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7.3 POST-TUTORIAL

1. Illustrate the complexity of the steps involved in Wolfe's method.

Solution:

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	Marks Secured ..... out of 50
	Full Name of the Evaluator:
	Signature of the Evaluator:
	Date of Evaluation:

Tutorial 8

Quadratic Optimization using Beale's Method

Date of the Session: .....

Learning outcomes:

- Understanding Kuhn tucker conditions
- Understanding the process of Beale's method

8.1 PRE-TUTORIAL

1. Outline the steps involved in Beale's method.

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8.2 IN-TUTORIAL

1. Use Beale's method for solving quadratic programming problem.

$$\begin{aligned}
&\text{Maximize:} \quad Z = 4x_1 + 6x_2 - 2x_1^2 - 2x_1x_2 - 2x_2^2 \\
&\text{Subject to:} \\
&\quad x_1 + 2x_2 \leq 2 \\
&\quad x_1, x_2 \geq 0
\end{aligned}$$

Solution:

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8.3 POST-TUTORIAL

1. Compute the first iteration and identify the entering and leaving variables:

$$\begin{aligned}
&\text{Minmize:} \quad Z = -4x_1 - 6x_2 + x_1^2 + 2x_2^4 \\
&\text{Subject to:} \\
&\quad -x_1 - 2x_2 \geq -4 \\
&\quad x_1, x_2 \geq 0
\end{aligned}$$

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Tutorial 9

Geometric Programming

Date of the Session: .....

Learning outcomes:

1. Introduce Posynomials, and arithmetic mean - geometric mean inequality.

2. Teach optimization of posynomial objective functions using Geometric Programming.

3. Introduce the concept of degree of difficulty.

4. Sketch the method of solving problem with one degree of difficulty.

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9.1 PRE-TUTORIAL

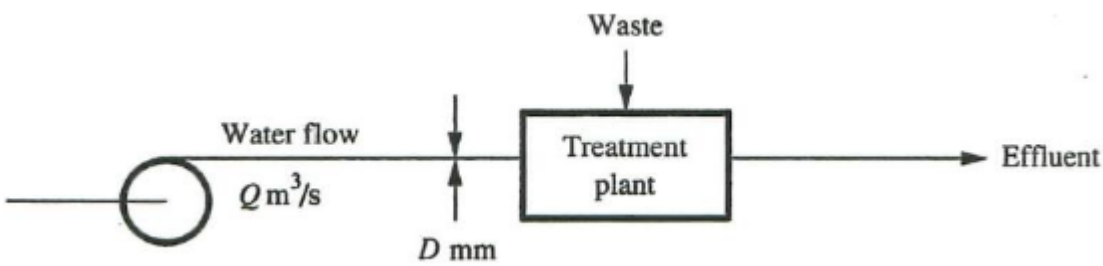
1. How to compute the degree of difficulty of Geometric programming problem?

2. List the areas of application of Geometric programming.

9.2 IN-TUTORIAL

1. Determine the optimum pipe diameter which results in minimum first plus operating cost for 100 m of pipe. The objective function is to determine the optimum pipe diameter for highest discharge.

$$y = 160D + \frac{32 * 10^{12}}{D^5}$$



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9.3 Case Study

9.3.1 Digital Marketing Campaign

Consider a company that wants to allocate its advertising budget across different digital marketing channels, such as social media, search engine marketing (SEM), and display advertising. The goal is to maximize the overall reach and engagement while staying within the budget constraints. **Advertising Goals:**

Table 9.1: Channel Information

Channel Cost per Click ($)	Click-Through Rate (%)	Budget Allocation ($)
Social Media 5000	0.50	3
SEM 7000	0.75	2
Display Advertising 4000	0.30	4

- Minimize the total cost.
- Achieve a minimum combined click-through rate of 5%.

Write Python code to solve this advertising budget allocation problem using Geometric Programming.
Solution:

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9.3.2 Product Launch Advertising

Consider a company launching a new product and planning its advertising budget across online and offline channels, such as social media, television, and print. The goal is to maximize the product's visibility while staying within the budget constraints. **Advertising Goals**

Table 9.2: Channel Information

Channel	Cost per Impression ($)	Impressions per Dollar	Budget Allocation ($)
Social Media	0.10	8	10000
Television	0.30	4	8000
Print	0.20	6	5000

- Minimize the total cost.
- Achieve a minimum combined impressions per dollar of 25.

Write Python code to solve this advertising budget allocation problem using Geometric Programming.
Solution:

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Tutorial 10

Genetic Algorithm (GA)

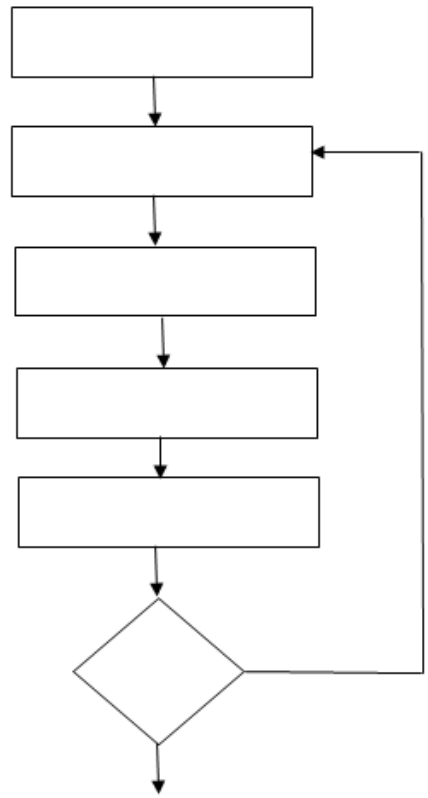
Date of the Session: .....

Learning outcomes:

- Understanding the concept on simulated annealing.
- Solving problems on simulated annealing.

10.1 PRE-TUTORIAL

1. Name and describe the main features of Genetic Algorithms (GA) and the also arrange the features of Genetic Algorithms in the flowchart given below.



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2. Suppose a genetic algorithm uses chromosomes of the form $x = abcdefgh$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) = (a + b) - (c + d) + (e + f) - (g + h)$ and let the initial population consist of four individuals with the following chromosomes:

$$\begin{aligned} x_1 &= 6\ 5\ 4\ 1\ 3\ 5\ 3\ 2 \\ x_2 &= 8\ 7\ 1\ 2\ 6\ 6\ 0\ 1 \\ x_3 &= 2\ 3\ 9\ 2\ 1\ 2\ 8\ 5 \\ x_4 &= 4\ 1\ 8\ 5\ 2\ 0\ 9\ 4 \end{aligned}$$

Maximize the above function.

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10.2 IN-TUTORIAL

1. Maximize the value of the function
$$f(x) = -x^2 + 2x$$
over the range of real numbers 0 to 2, with initial population 111010, 01010, 10101, 00101 with random numbers 0.3, 0.2, 1.3, 0.9. Crossover point can be taken randomly.

Solution:

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10.3 Case Study

10.3.1 Machine Scheduling in a Manufacturing Plant

Consider a manufacturing plant with multiple machines and a set of jobs to be processed on these machines. Each job has a processing time, and each machine has a different processing speed. The goal is to find an optimal schedule that minimizes the total completion time for all jobs.
Machine and Job Information: Machines: M1, M2, M3
Jobs: J1, J2, J3, J4, J5
Processing Times:
J1: 4, J2: 5, J3: 2, J4: 7, J5: 3
Processing Speeds: M1: 2, M2: 3, M3: 1 **Objective:** Minimize the total completion time for all jobs. Write Python code to solve this machine scheduling problem using a Genetic Algorithm **Solution**:

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10.3.2 Project Task Scheduling with Dependencies

Consider a project with multiple tasks that need to be scheduled. Each task has a duration and dependencies on other tasks. The goal is to find an optimal schedule that minimizes the project completion time while satisfying task dependencies.
Task Information:
Tasks: A, B, C, D, E
Duration: A: 2, B: 3, C: 4, D: 2, E: 5 **Dependencies:** B depends on A
C depends on A
D depends on B, C
E depends on D
Objective: Minimize the total project duration. Write Python code to solve this project task scheduling problem using a Genetic Algorithm. **Solution**:

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Tutorial 11

Ant Colony Optimization (ACO)

Date of the Session: .....

Learning outcomes:

- Understanding different approaches for solving problems of Ant Colony Optimization (ACO)
- Applying various approach to solve Ant Colony Optimization (ACO).

11.1 PRE-TUTORIAL

1. What is swarm intelligence?

2. What is meant by Ant Colony Optimization?

3. State the Merits and demerits of Ant Colony Optimization.

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11.2 IN-TUTORIAL

1. An ANT is at a distance of 5m from the TREE 15m from CAR and 4m from a DOLL, the distance between TREE and the CAR is 4m, CAR and DOLL is 1m, DOLL and TREE is 8m. Solve using Ant Colony Optimization to find the shortest route to travel through tree, car and doll.

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11.3 POST-TUTORIAL

1. Suppose we have a TSP problem with 5 cities. The distance between cities is given by the following matrix

$$d = \begin{bmatrix} 0 & 10 & 12 & 11 & 14 \\ 10 & 0 & 13 & 15 & 8 \\ 12 & 13 & 0 & 9 & 14 \\ 11 & 15 & 9 & 0 & 16 \\ 14 & 8 & 14 & 16 & 0 \\ 14 & 8 & 14 & 16 & 0 \end{bmatrix}$$

   Determine the best route with minimum total distance by using the ACO.

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Tutorial 12

Particle Swarm Optimization (PSO)

Date of the Session: .....

Learning outcomes:

- Understanding the concept of Particle Swarm Optimization (PSO).
- Apply Particle Swarm Optimization in real-time situations.

12.1 PRE-TUTORIAL

1. What is a particle swarm and their behavior under the PSO algorithm?

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2. What are the variations and limitations of PSO algorithm?

3. Write down the steps involved in solving the problem using particle swarm optimization?

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12.2 IN-TUTORIAL

1. Find out the maximum of

$$\text{F(X)} \quad = -x^2 + 5x + 20 \text{ with } -10 \leq x \leq 10$$

Using the PSO algorithm. Use 9 particles with the initial positions $x_1 = -9.6$, $x_2 = -6$, $x_3 = -2.6$, $x_4 = -1.1$, $x_5 = 0.6$, $x_6 = 2.3$, $x_7 = 2.8$, $x_8 = 8.3$, $x_9 = 10$. Show the detailed computations for iterations 1,2 and 3.
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2. Illustrate the steps involved in Particle Swarm Optimization

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12.3 POST-TUTORIAL

1. Use Particle Swarm Optimization technique (PSO), and solve the following problem .

$$\text{Maximize:} \quad f(x) = x_1^2 - x_1x_2 + x_2^2 + 2x_1 + 4x_2 + 3 \text{ with } -5 \leq x_1, x_2 \leq 5$$

	x_1	x_2
1	-3	2
2	1	4
3	-2	-4
4	3	-2

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